RAGUNATH PERIYASAMY

Robotics Software & AI Engineer

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PROFESSIONAL SUMMARY

Research Engineer at RoboMiracle building software for service robots deployed in hospitals, schools and workplaces across India — covering **navigation & autonomy**, **touchscreen UI**, **voice & AI interaction**, and **backend & hardware integration**, through to on-site installation. Background in Electronics & Communication Engineering with hands-on experience in embedded vision, full-stack development and applied AI, and a strong interest in on-device AI, LLMs and computer vision.

EXPERIENCE

Research Engineer | *RoboMiracle, Coimbatore* Aug 2026 – Present

- Build robot software across **navigation & autonomy**, **touchscreen UI & apps**, **voice & AI interaction** and **backend & hardware integration** using Python, ROS / ROS 2, Raspberry Pi and NVIDIA Jetson.
- **Nexus A1** — service humanoid with touchscreen, deployed at a hospital in Thiruvananthapuram: robot software and on-site installation.
- **Medi Buddy** — autonomous hospital logistics robot moving sterile supplies to operating theatres at Aravind Eye Care: navigation and UI software.
- **Nova A1** — AI educational humanoid for schools and STEM labs with face recognition and voice commands: voice and AI interaction.
- **Aurra** — humanoid robot for workplaces and events: robot software. Also contributed to **Nila** (reception) and **Dine Bot** (restaurant service) robots.

Software Engineer | *dsignz media, Coimbatore* Jun 2025 – Aug 2026

- Contributed to a Laravel-based internal Workforce Management admin dashboard used for employee and project tracking.
- Developed full-stack features across the Python and PHP/Laravel stack.
- Built and deployed AI-integrated tools, including RAG-based systems and OCR pipelines.

PROJECTS

Mister Mini — 3D-Printed Humanoid Robot

Adaptive Autonomous Assistance using Raspberry Pi · College final-year project, team of 6

- Built a humanoid robot on a **Raspberry Pi 4B** with an **OAK-D Lite** depth camera and servo-driven head, arms and hands.
- Implemented real-time **face recognition and emotion detection** with on-device inference, and designed the vision-to-actuation pipeline from camera input to servo response.

Face Recognition IoT System

- OAK-D Lite face recognition system with a Tkinter desktop GUI for real-time monitoring; integrated **Bolt IoT** and the **Telegram API** to send remote alerts on recognition events.

RAG Chatbot from Scratch

- Retrieval-augmented generation chatbot built from the ground up: document chunking and embedding, vector retrieval of relevant passages, and LLM answers grounded in them.

TECHNICAL SKILLS

- **Robotics & Hardware:** ROS / ROS 2, robot navigation, Raspberry Pi 4B, NVIDIA Jetson, OAK-D Lite, servo control, GPIO & sensor interfacing, 3D printing
- **AI & Computer Vision:** Face recognition, emotion detection, OCR, LLMs, RAG, FAISS, Ollama (local LLMs)
- **Programming:** Python, PHP (Laravel)
- **Backend & Data:** Flask, MySQL, SQLite
- **Interfaces & IoT:** Touchscreen UI, Tkinter, Streamlit, Bolt IoT, Telegram API
- **Tools:** Git

EDUCATION

B.E., Electronics & Communication Engineering | *Muthayammal Engineering College* Graduated 2025